

Energy Autonomy and Human Well-Being in Greek Buildings: An Integrated Parametric, Multi-Criteria and Surrogate-Model Framework Applied to 69 Locations

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Abstract

Early building design in Greece still treats energy-demand reduction, on-site generation and occupant well-being as separate problems, solved late and by different specialists. This paper proposes an integrated framework that couples parametric design, hourly building-performance simulation, energy-balance and storage-dispatch algorithms, a computable well-being indicator set and AI-based surrogate models into one early-stage workflow. Energy autonomy is defined temporally, through hourly self-sufficiency and self-consumption, rather than as an annual net balance. Well-being is expressed through thermal, visual, circadian, air-quality and view indicators derived from EN 16798-1, EN 17037 and CIE S 026, complemented by a transparent, user-weighted qualitative layer. The framework is contextualised for Greece's four KENAK climate zones the 2024 transition from net metering to net billing and the 2024 EPBD recast, which makes zero-emission buildings, on-site solar and indoor environmental quality joint objectives. It is demonstrated with a reduced-order hourly model applied to 69 locations spanning Crete, the Dodecanese, the Cyclades, the North and East Aegean, the Sporades, the Ionian Islands and all mainland regions. The synthetic climate reproduces published heating degree days of the zone reference cities within -8% to $+3\%$. For a 450 m^2 residential reference building, annual photovoltaic generation exceeds annual demand everywhere (125–173%), yet hourly self-sufficiency is only 47–66%, exposing a large gap between net-zero and autonomy. Battery capacity and roof PV dominate autonomy, whereas glazing ratio, insulation and form dominate well-being; the objectives conflict only near the Pareto front, where maximising autonomy costs 13–21 well-being points. Under a $+2\text{ }^{\circ}\text{C}$ scenario, autonomy improves slightly while passive comfort in southern Zone A locations declines by up to 15 percentage points. A cross-city surrogate predicts 14 unseen locations with $R^2 = 0.96$ for self-sufficiency and 0.89 for well-being. The methods are released as an open web tool covering all 69 locations. All numerical results are indicative and require validation with EnergyPlus and Radiance.

Keywords: energy autonomy; human well-being; building-integrated photovoltaics; battery storage; parametric design; surrogate modelling; regenerative design; KENAK; EPBD 2024; net billing; Greek islands

1 Introduction

Buildings that generate and store their own energy while supporting the health of their occupants are central to a regenerative built environment (Reed, 2007; Cole, 2012). Yet the decisions that most constrain both outcomes—massing, orientation, window-to-wall ratio, shading and the surface area given to photovoltaics—are taken in early design, when designers have the least analytical support (Østergård et al., 2016).

Greece makes this problem sharper. Solar resource is abundant, so building-integrated photovoltaics (BIPV) can cover a large share of annual demand. However, cooling demand peaks on summer afternoons, while heating and lighting demand peaks on winter evenings, so an annual surplus rarely translates into hour-by-hour autonomy. The existing stock is old and poorly insulated: according to the 2011 census, 45.6% of Greek dwellings had no insulation at all, and buildings constructed before the 1979 Thermal Insulation Regulation account for about 45–46% of floor area (Touloupaki and Theodosiou, 2017). Air-conditioning ownership is among the highest in Europe, at roughly two-thirds of households (ICE, 2026), and the Mediterranean is a recognised climate-change hot spot (Giorgi, 2006). Projections for Greece show heating degree days falling and cooling degree days rising, with the largest absolute cooling increases in the three summer months (Karagiannidis et al., 2024).

European policy now points in the same direction. The recast Energy Performance of Buildings Directive (EU) 2024/1275 requires all new buildings to be zero-emission from 2030 (public buildings from 2028), phases in mandatory solar installations, and, for the first time, defines indoor environmental quality in EU law and asks Member States to set requirements for a healthy indoor climate (EUR-Lex, 2024; BUILD UP, 2025). Energy autonomy and occupant well-being are therefore becoming joint regulatory objectives, not optional extras.

Two further Greek specificities motivate this work. First, the national self-consumption framework moved from net metering to net billing in 2024 (Tsagas, 2025; Lexology, 2024), which makes the timing of generation and consumption economically relevant for the first time. Second, dozens of Aegean islands are non-interconnected or weakly connected; projects such as the TILOS hybrid microgrid show that local autonomy is both feasible and valued (CORDIS, European Commission, 2018; Tsagas, 2024).

1.1 Research gaps

1. **Separation of demand and supply.** Passive design and on-site generation are usually optimised in sequence: the facade is designed first and PV is added to leftover surfaces.
2. **Annual rather than temporal balance.** Net-zero metrics hide the mismatch between generation and load that determines storage needs and grid dependence (Sartori et al., 2012; Salom et al., 2011).
3. **A reduced notion of well-being.** Well-being is usually collapsed into a comfort threshold, ignoring circadian light, views, air quality and qualitative criteria.
4. **Computational cost.** Coupled thermal, daylight and PV simulation is too slow for interactive exploration of thousands of alternatives (Westermann and Evins, 2019).
5. **Geographic coverage.** Greek studies typically use one reference city per KENAK zone, although islands and mainland locations within the same zone differ considerably in climate and grid context.

1.2 Objectives and contributions

The research pursues three objectives. **O1** develops parametric workflows linking form, passive strategies, use patterns, current and future climates, and dynamic energy balance with BIPV, storage and grid interaction. **O2** develops multi-criteria methods combining energy autonomy with quantitative and qualitative well-being indicators. **O3** trains and validates surrogate models and embeds them in tools for real-time comparison.

This paper contributes (i) a framework that treats envelope area as contested between generation, passive performance and well-being; (ii) a computable well-being indicator set separating physics-based metrics from value-weighted criteria; (iii) a hybrid physics–ML surrogate strategy that learns load and generation behaviour but keeps storage dispatch physical; (iv) a national demonstration over 69 Greek locations, including 31 island locations; and (v) an open web tool implementing the demonstrator.

Section 2 reviews the state of the art, Section 3 the Greek context, Section 4 the methodology, Section 5 the demonstrator, Section 6 the results, and Sections 7–8 discussion and conclusions.

2 Background and state of the art

2.1 From green to regenerative design

Regenerative design shifts the goal from doing less harm to producing net benefits for people and ecosystems (Reed, 2007; Cole, 2012). Certification schemes operationalise this partly: the Living Building Challenge requires net-positive energy and addresses health and biophilia, while WELL sets targets for air, light, thermal comfort and mind. Both remain largely checklist-based and are rarely usable as continuous design objectives, which motivates the computable indicators in Section 4.3.

2.2 Energy balance and autonomy metrics

The net-zero energy building literature established that an annual balance is insufficient and that load matching and grid interaction must be assessed at short time steps (Sartori et al., 2012; Salom et al., 2011). Self-consumption (SC) and self-sufficiency (SS) are now the standard pair of indicators for PV buildings (Luthander et al., 2015):

$$SC = \frac{\sum_t \min(G_t, L_t) + E^{ch}}{\sum_t G_t}, \quad SS = \frac{\sum_t L_t - \sum_t I_t}{\sum_t L_t}, \quad (1)$$

where G_t is on-site generation, L_t electrical load, I_t grid import in hour t , and E^{ch} the PV energy charged into the battery. Storage autonomy (hours of average night load a full battery covers) and a grid-interaction index complete the set.

2.3 BIPV and storage in design

BIPV turns envelope surfaces into generators, but facade modules produce less than optimally tilted roof arrays and change the thermal and daylight behaviour of the facade. Most design studies still size PV after the architecture is fixed. Storage adds a temporal dimension whose value depends on the shape of load and generation profiles, not on annual totals.

2.4 Well-being in building performance

Thermal comfort is codified through the PMV/PPD model and the adaptive model (de Dear and Brager, 1998; Nicol and Humphreys, 2002), adopted in EN 16798-1 (CEN, 2019). Climate-based daylight metrics (Reinhart et al., 2006) and glare probability (Wienold and Christoffersen, 2006) underpin EN 17037 (CEN, 2018). Research on the non-visual effects of light (Lucas et al., 2014) led to melanopic metrics in CIE S 026 (CIE, 2018). Qualitative dimensions such as thermal delight (Heschong, 1979), view quality and biophilic connection are recognised but rarely computed in design tools.

2.5 Surrogate modelling for building design

Surrogate models replace expensive simulation with fast statistical approximations and are increasingly used in early design (Westermann and Evins, 2019; Østergård et al., 2016). Gradient-boosted trees, Gaussian processes and neural networks are common. Most surrogates predict scalar annual outputs, which cannot support temporal autonomy analysis, and few combine energy and multi-domain well-being targets or generalise across climates.

2.6 Energy autonomy on Greek islands

The TILOS project installed an 800 kW wind turbine, a 160 kW PV park and a 2.4 MWh battery on a Dodecanese island of about 500 permanent residents (CORDIS, European Commission, 2018); the island now retains supply when the subsea link or the diesel plant on Kos fails (Tsagas, 2024). Astypalaia, Symi and Kastellorizo were selected as pilot “smart islands” because they are not scheduled for interconnection (Norton Rose Fulbright, n.d.). These projects work at island scale; the present framework addresses the complementary building scale.

3 The Greek context

3.1 Climate zones and regulation

The Regulation on the Energy Performance of Buildings (KENAK), revised in 2017, sets envelope and system requirements by climate zone (EPBD Concerted Action, 2020; Dascalaki et al., 2012). Zones are defined by heating degree days (HDD): A (601–1100), B (1101–1600), C (1601–2200) and D (2201–2620) (Touloupaki and Theodosiou, 2017). Zone membership is assigned by prefecture (Technical Chamber of Greece, 2010). Zone A comprises Crete, the Cyclades, the Dodecanese, Samos, Messinia, Laconia, Argolida, Zakynthos, Kefalonia and Ithaca, Kythira and the Saronic islands, and lowland Arcadia. Zone B includes Attica, most of central and western Greece, Magnesia, Lesbos, Chios, Corfu, Lefkada, Thesprotia, Preveza and Arta. Zone C covers most of Thessaly, Epirus-Ioannina, central and eastern Macedonia, Thrace and mountainous Arcadia. The 2017 revision moved Drama to Zone D, which also contains Florina, Kozani, Kastoria, Grevena and north-eastern Serres (Panousis, 2018). Figure 1 shows the 69 locations used here.

3.2 Self-consumption framework

Law 5106/2024 closed net metering to most consumers, and net billing became the dominant self-consumption scheme from September 2024 (Tsagas, 2025; Lexology, 2024). Exported surplus is valued at wholesale-linked prices, and on non-interconnected islands at a special market price



Figure 1: The 69 study locations coloured by KENAK climate zone (27 in Zone A, 20 in B, 17 in C and 5 in D), including 31 island locations.

(Clean Energy for EU Islands, 2025). Under net metering, annual net-zero was economically sufficient; under net billing, a kilowatt-hour exported at noon is worth less than one imported in the evening. Self-consumption and self-sufficiency therefore become economic as well as environmental objectives.

3.3 The 2024 EPBD recast

Directive (EU) 2024/1275 entered into force in May 2024 and had to be transposed by May 2026. Three provisions matter directly for this framework. First, all new buildings owned by public bodies must be zero-emission buildings (ZEB) from 1 January 2028 and all new buildings from 1 January 2030; a ZEB has very low energy demand, no on-site fossil emissions and very low operational emissions (EUR-Lex, 2024). Second, Article 10 phases in suitable solar installations: on new non-residential buildings from 31 December 2026 and on new residential buildings from 31 December 2029 (Longevity Partners, 2026). Third, the recast defines indoor environmental quality as an assessment of conditions affecting occupants' health and well-being, based on parameters such as temperature, humidity, ventilation rate and contaminants (BUILD UP, 2025); Article 13 asks Member States to set requirements for a healthy indoor climate, and non-residential ZEBs must be equipped with IEQ monitoring and control (AIVC, 2026). Life-cycle global warming potential must also be calculated, first for large new buildings and for all new buildings by 2030 (EUR-Lex, 2024).

For Greek designers, this means that PV on new buildings will become the norm, so the question shifts from *whether* to install PV to *how* to use it for autonomy, and that energy measures must be demonstrably compatible with indoor environmental quality. Both questions are central

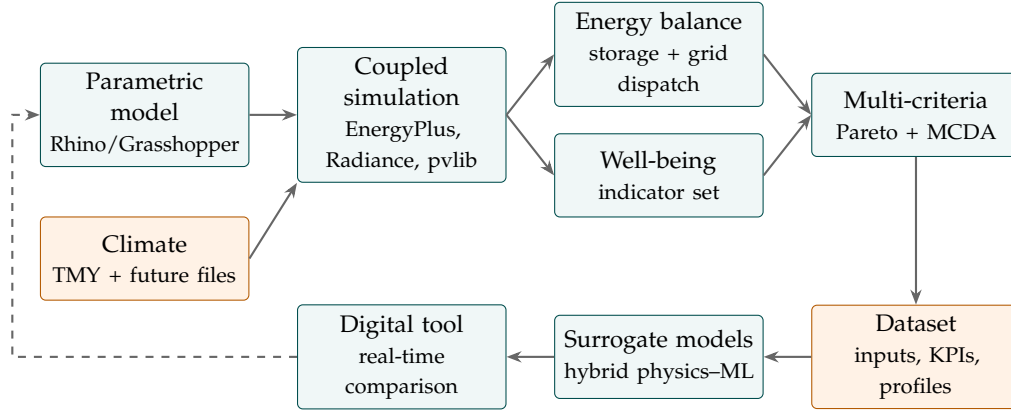


Figure 2: The integrated framework. Solid arrows show data flow; the dashed arrow shows the designer’s iteration loop enabled by the surrogate-based tool.

to the framework proposed here.

3.4 Heat, climate change and well-being

Degree-day projections show decreasing HDD and increasing CDD across Greece, with elevation and sea proximity shaping the pattern (Karagiannidis et al., 2024). For tourist islands, thermal-stress studies such as that for Santorini point to growing summer discomfort (Katavoutas et al., 2021). Future weather files for Greek locations should be generated by morphing typical years with climate-model projections (Jentsch et al., 2008; Rodrigues et al., 2023).

4 Methodology

The framework (Figure 2) links six stages: parametric model generation, coupled simulation, energy-balance and dispatch, well-being evaluation, multi-criteria analysis, and surrogate-based tools.

4.1 Parametric workflow (O1)

Geometry is generated in Rhino/Grasshopper and translated through Ladybug Tools (Roudsari and Pak, 2013) into EnergyPlus (Crawley et al., 2001) for thermal loads and Radiance for daylight and facade irradiance. PV yield is computed per surface with pvlib (Holmgren et al., 2018), including cell temperature and partial shading. Table 1 lists the design space.

4.2 Energy balance and dispatch (O1)

Hourly electrical load—heat pump, lighting, equipment and domestic hot water—is balanced against PV generation. A rule-based strategy serves the load, then charges the battery, then exports; an optimisation-based strategy (MILP or model-predictive control) provides the upper bound of achievable autonomy. Battery state of charge evolves as

$$SOC_{t+1} = SOC_t + \eta_c P_t^{ch} \Delta t - \frac{P_t^{dis} \Delta t}{\eta_d}, \quad SOC_{min} \leq SOC_t \leq SOC_{max}. \quad (2)$$

Outputs are SS, SC, annual import and export, peak import, storage autonomy and the net-billing cost.

Table 1: Parametric design variables and ranges used in the demonstrator.

Group	Parameters	Range
Form	Length:depth ratio; orientation of main facade	0.6–3.0; $\pm 60^\circ$ from south
Envelope	Main- and other-facade glazing ratio; glass g-value; insulation level; thermal mass	10–80%, 10–50%; 0.25–0.65; KENAK / –40% / nZEB+; light–heavy
Passive	External overhang depth; night ventilation	0–1.2 m; on/off
Generation	Roof PV coverage; facade BIPV on opaque main facade	20–100%; 0–100%
Storage	Battery capacity	0–40 kWh (450 m ²)
Use	Residential, office, seasonal hotel (April–October)	schedules

4.3 Computable well-being indicators (O2)

Table 2 lists the indicator set. Physics-based indicators are normalised to 0–100 against their targets; qualitative scores remain a separate, visible layer so that value judgements are never hidden inside physical metrics. The well-being index is a weighted mean,

$$WB = \frac{\sum_i w_i s_i}{\sum_i w_i}, \quad (3)$$

with weights w_i elicited from stakeholders; this paper uses thermal 3, daylight 2 and 1 for glare, circadian light, view and fresh-air hours.

4.4 Multi-criteria analysis (O2)

Multi-objective optimisation with NSGA-II (Deb et al., 2002) produces Pareto fronts between autonomy and well-being. TOPSIS or weighted ranking with user-defined weights then supports decisions, followed by a weight-sensitivity analysis.

4.5 Dataset generation and surrogate models (O3)

Designs are sampled by Latin hypercube; global sensitivity analysis (Herman and Usher, 2017) prunes weak parameters before denser sampling. The surrogate strategy is hybrid: machine learning predicts load and PV behaviour (in the full framework, hourly profiles or their principal components), while the fast dispatch algorithm runs on the predictions, keeping battery size and tariff as free inputs. Validation uses held-out designs and held-out locations, with R^2 , RMSE and CV(RMSE) reported against ASHRAE Guideline 14 (ASHRAE, 2014). SHAP values explain predictions (Lundberg and Lee, 2017).

Table 2: Computable well-being indicators, their full-framework definitions and the proxies used in the demonstrator.

Domain	Full framework	Demonstrator proxy
Thermal	Occupied hours within adaptive Category II (CEN, 2019); TM52 overheating (CIBSE, 2013)	Free-running hours within the adaptive range
Visual	sDA _{300/50%} , ASE; EN 17037 levels	Daylit floor fraction from average daylight factor (Littlefair, 2011)
Glare	Hours with DGP > 0.40 (Wienold and Christoffersen, 2006)	Low-sun beam exposure weighted by glazed area
Circadian	Morning melanopic EDI $\geq$ 250 lx (CIE, 2018)	Days with ≥ 3 morning hours above threshold
Air	Occupied hours with CO ₂ above Category II	Hours mild enough for window ventilation
View	EN 17037 view-out level	Level from main-facade glazing ratio
Qualitative	Thermal delight, biophilia, control, spatial variety	1–5 team scores (web tool only)

5 Demonstrator and case-study set-up

5.1 Locations and synthetic climate

The demonstrator covers 69 locations (Figure 1, Appendix A). For each, approximate long-term monthly mean temperatures were derived from January and July normals and a regional seasonal shape (island, coastal, inland, mountain); daily temperature ranges vary by the same classes. Monthly global horizontal irradiation was interpolated by latitude between southern and northern reference profiles, with adjustments for islands, mountains and the wetter Ionian winter. Hourly weather is synthesised with seeded day-to-day clearness and temperature variability; solar position, Erbs diffuse splitting (Erbs et al., 1982) and isotropic plane-of-array irradiance follow. Annual irradiation ranges from 1,584 to 1,893 kWh/m² (Figure 3).

Figure 3 shows that temperature varies far more across Greece than solar resource does, and that distance from the sea matters as much as latitude. In the north-west, Corfu and Ioannina lie only about 80 km apart, yet Ioannina's January mean is about 5 °C lower (4.9 against 10.1 °C) and its heating degree days are some 70% higher, while its irradiation is almost identical. The same contrast appears in the Peloponnese between mountainous Tripoli and coastal Kalamata. The Aegean islands have the smallest annual temperature ranges (about 13–15 K in Santorini, Rhodes and Heraklion, against 22 K in Larissa), which reduces both heating and extreme cooling loads. Among the islands, Lemnos in the north Aegean is markedly cooler in winter than the Cyclades or the Dodecanese. Irradiation differs by at most about 20% between the sunniest (Crete, Dodecanese) and least sunny (western Macedonia) locations, so differences in autonomy across Greece are driven mainly by the timing and size of heating and cooling demand rather than by the solar resource.

Table 3 compares synthetic HDD (base 18 °C) with values published for the KENAK zone reference cities (Touloupaki and Theodosiou, 2017); agreement is within −8% to +3%.

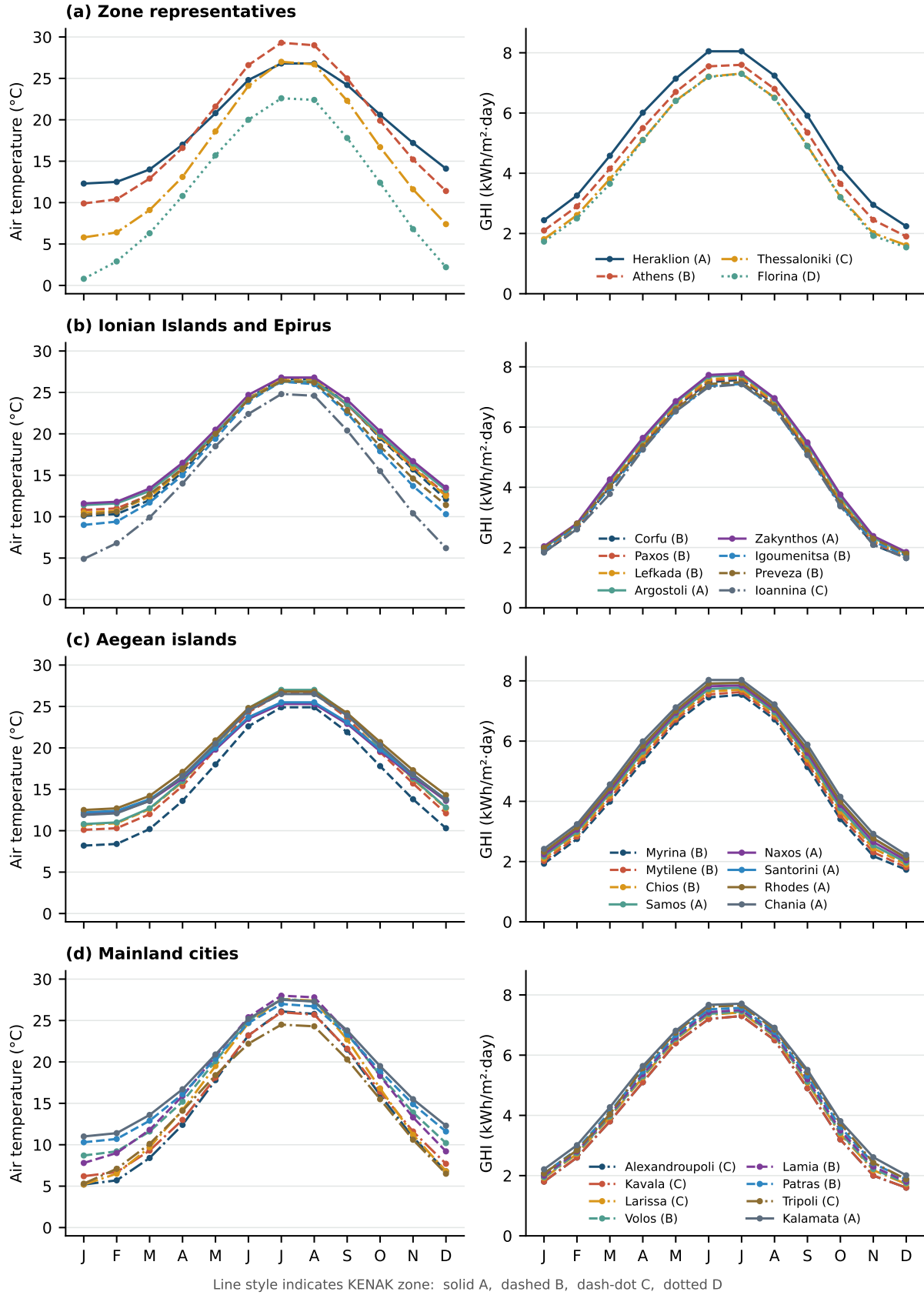


Figure 3: Monthly mean air temperature (left) and global horizontal irradiation (right) used by the demonstrator: (a) the four KENAK zone representatives; (b) the Ionian Islands and Epirus, including Corfu, Igoumenitsa and Ioannina; (c) Aegean islands from Lemnos to Crete; (d) mainland cities from Thrace to the Peloponnese. Line style indicates the KENAK zone.

Table 3: Validation of synthetic heating degree days against published values.

City	Zone	Published HDD	Synthetic HDD	Difference
Heraklion	A	702	645	−8%
Athens	B	947	958	+1%
Thessaloniki	C	1677	1731	+3%
Kastoria	D	2420	2443	+1%

5.2 Thermal and PV model

A single-zone 1R1C model (ISO 13790 capacitance classes, [ISO, 2008](#)) is solved with an implicit hourly step,

$$T_{i,t+1} = \frac{\frac{C}{\Delta t} T_{i,t} + Q_{sol} + Q_{int} + (H_{tr} + H_{ve})T_e + H_g T_g + Q_{hvac}}{\frac{C}{\Delta t} + H_{tr} + H_{ve} + H_g}, \quad (4)$$

run twice: with ideal heat pumps (temperature-dependent COP and EER) to compute energy, and free-running with window and night ventilation to compute passive comfort. Overhang shading uses the solar profile angle. PV includes a temperature correction and a performance ratio of 0.86. The 2050 case applies +1.8 °C (winter) to +2.5 °C (summer), an illustrative shift rather than a morphed climate-model file.

5.3 Reference building

The reference is a 450 m², three-storey residential building with a length:depth ratio of 1.8, a south main facade with 40% glazing (20% elsewhere), a 0.6 m overhang, insulation 40% below approximate KENAK limits, heavy mass, night ventilation and heat recovery, 80% roof PV, 50% facade BIPV on the opaque main facade (28.4 kWp) and a 15 kWh battery.

6 Results

6.1 National patterns of autonomy and comfort

The reference building reaches 47–66% self-sufficiency across the 69 locations (Figure 4). Zone means are 64% (A), 61% (B), 54% (C) and 49% (D). Island locations average 63% against 56% on the mainland, because milder winters reduce the evening heating load that PV cannot cover. Passive comfort follows a similar gradient, from about 94% of occupied hours in Crete and the Dodecanese to about 60% in Macedonia and Thrace (Table 4).

6.2 The gap between net-zero and autonomy

Annual PV generation exceeds annual demand at every location, by 25–73% (Figure 5). Nevertheless, 34–53% of demand is still imported. The locations with the greatest annual surplus (Crete, the Dodecanese) gain the least additional self-sufficiency from it, because the surplus is concentrated at summer midday.

6.3 Temporal structure

The hourly balance maps (Figure 6) show why. Every location has a solar surplus from mid-morning to mid-afternoon in all months, but a deficit every night; in Zones C and D, winter

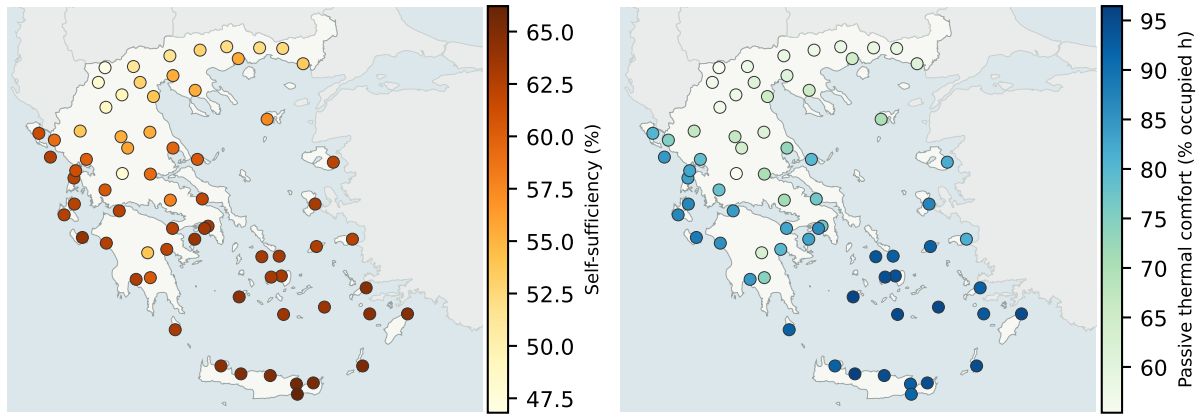


Figure 4: Reference-design results across Greece: hourly self-sufficiency (left) and passive thermal comfort (right).

Table 4: Regional summary for the reference design, sorted by heating degree days. Q_h , Q_c : heating and cooling need; PC: passive comfort; ΔPC_{2050} : change under the warming scenario.

Region	n	Zones	HDD (K d)	CDD	SS (%)	Q_h (kWh/m ² yr)	Q_c	PC (%)	ΔPC_{2050} (pp)
Crete	6	A	610	1142	65	2	30	94	-9
Dodecanese	5	A	653	1090	64	3	28	94	-7
Cyclades	6	A	734	905	63	3	20	94	-2
Saronic Islands	1	A	796	1252	64	5	36	83	-6
Ionian Islands	7	AB	865	992	63	6	25	86	+0
Attica	2	B	905	1261	63	6	35	82	-4
North & East Aegean	5	AB	994	952	62	8	24	83	+2
Peloponnese	7	ABC	1056	1033	61	9	25	79	+0
Sporades	1	B	1085	846	61	9	20	80	+7
Western Greece	1	B	1090	1055	60	9	24	78	+5
Epirus	4	BC	1262	867	59	12	19	76	+5
Central Greece	4	BC	1520	867	57	17	20	68	+1
Thessaly	4	BC	1553	962	57	18	23	66	+1
Thrace	3	C	1929	759	53	25	16	60	+3
Macedonia	13	CD	2038	652	52	26	12	60	+4

mornings and evenings add large heating deficits. Monthly totals (Figure 7) show that grid imports concentrate in December–February in Florina, whereas in Athens and Corfu they are more evenly distributed and partly driven by summer night-time cooling.

6.4 Design-space exploration and trade-offs

For six locations, 400 Latin-hypercube designs each were simulated (Figure 8). Self-sufficiency spans roughly 20–96% and the well-being index 35–90. The highest-autonomy designs use about 19% main-facade glazing, freeing facade area for BIPV; the highest-well-being designs use 57–67%. Across the full samples, the two objectives are almost uncorrelated, so the conflict appears only near the Pareto front, where the highest-autonomy design costs 13–21 well-being points relative to the best.

Standardised regression coefficients pooled over the six locations (Figure 9) confirm that different parameters govern the two objectives. Battery capacity (0.62) and roof PV (0.42)

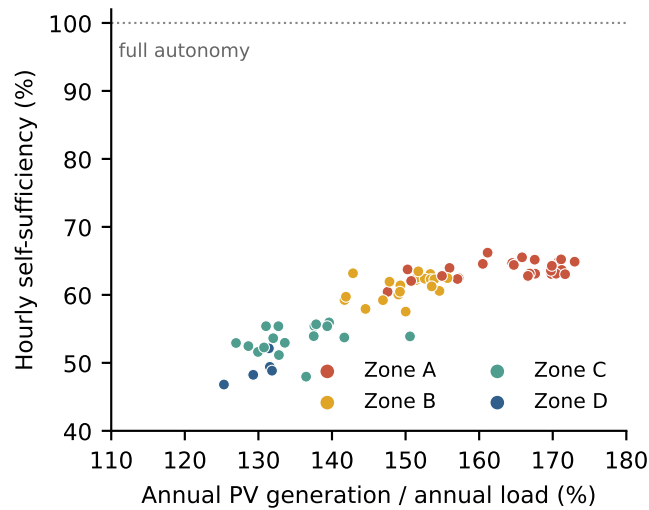


Figure 5: Annual PV-to-load ratio versus hourly self-sufficiency for all 69 locations.

dominate self-sufficiency, followed by insulation (0.20) and facade BIPV (0.17). Main-facade glazing (0.43), insulation (0.27), plan proportions (0.21) and thermal mass (0.15) dominate well-being. Main-facade glazing is the only strong parameter with opposite signs (-0.09 for autonomy, $+0.43$ for well-being), and insulation is the only parameter that strongly benefits both.

6.5 Use patterns

Occupancy timing changes autonomy as much as many design choices (Figure 10). With the same building, offices reach 65–84% self-sufficiency because their load coincides with daylight. Seasonal hotels, typical of Greek tourism, reach 63–72% because peak occupancy coincides with peak sunshine, against 47–65% for year-round residential use.

6.6 Climate change

Under the warming scenario, cooling need rises by $19 \text{ kWh/m}^2\text{yr}$ on average in Zone A and $10 \text{ kWh/m}^2\text{yr}$ in Zone D, while heating need falls by $3\text{--}8 \text{ kWh/m}^2\text{yr}$ (Figure 11). Self-sufficiency rises by 2.5–4.4 percentage points, because more load coincides with sunshine. Passive comfort moves in the opposite direction in the south: Zone A loses 4.3 points on average, and Ierapetra, Agios Nikolaos and Karpathos lose 9–15 points. In Zones C and D, milder winters improve passive comfort by 3–4 points. Autonomy and passive well-being can therefore diverge under climate change.

6.7 Surrogate generalisation across locations

A quadratic ridge-regression surrogate was trained on 60 designs from each of 55 locations (3,300 samples), using the 12 design parameters plus three climate descriptors (HDD, CDD, annual irradiation). Tested on 14 entirely unseen locations (840 samples), it predicted self-sufficiency with $R^2 = 0.96$ (RMSE 3.1 points) and the well-being index with $R^2 = 0.89$ (RMSE 2.6 points) (Figure 12). Even a simple surrogate can therefore extend a national dataset to new sites, which supports the proposed hybrid surrogate strategy.

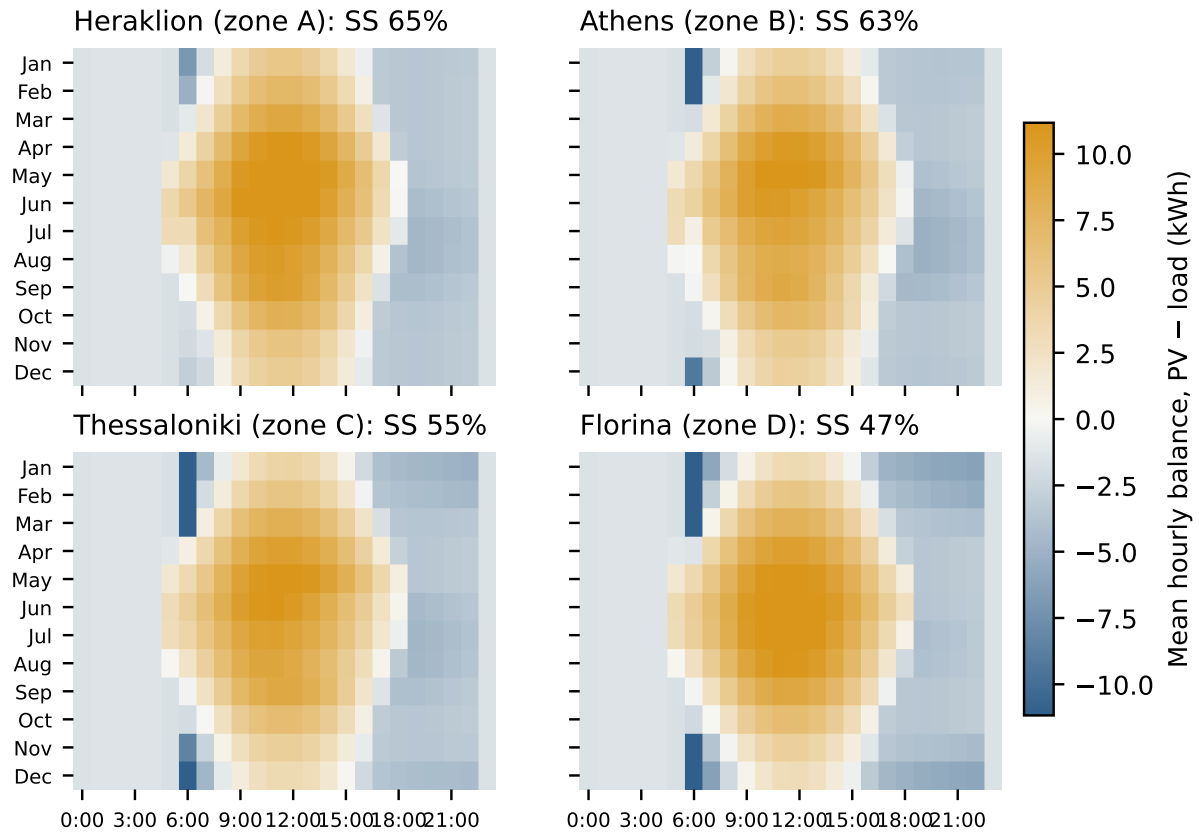


Figure 6: Mean hourly balance (PV minus load) for each month and hour in the four zone representatives.

7 Discussion

7.1 Implications for Greek design practice

Storage before more PV. Everywhere in Greece the reference building produces more electricity than it uses, yet imports a third to a half of its demand. Under net billing, storage or load shifting adds more autonomy per euro than further PV once annual generation exceeds load.

Facade BIPV is a well-being decision. Replacing glass with PV on the main facade raises autonomy only modestly when storage is limited, while reducing daylight, views and circadian light. Roofs should be saturated first.

Passive resilience matters most in the south. Crete, the Dodecanese and other Zone A locations lose passive comfort under warming even as autonomy improves. Shading, thermal mass and night ventilation protect occupants during outages and heatwaves, which is particularly relevant on islands with fragile grids.

Islands are natural early adopters. Island locations reach higher building-level autonomy, face higher electricity-system costs and have the most to gain from demand that follows local generation. Building-scale autonomy complements island-scale projects such as TILOS.

Supporting EPBD compliance. With solar becoming mandatory on new buildings and IEQ requirements entering national law, the framework offers a way to show that a design uses its mandatory PV well (self-consumption, self-sufficiency) and maintains a healthy indoor climate at the same time. Tools of this kind can support design teams during the transition from nZEB to ZEB requirements, alongside the official KENAK calculation.

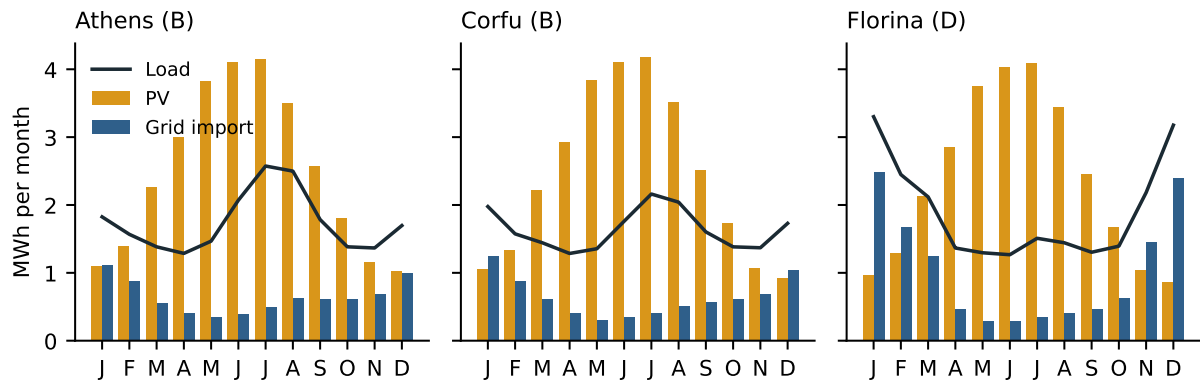


Figure 7: Monthly load, PV generation and grid import in Athens, Corfu and Florina.

Zones hide variation. Within Zone A, passive comfort ranges from 81% to 96% and within Zone B from 70% to 88%. Tools that expose location-specific performance, rather than zone averages, can guide better early decisions.

7.2 Limitations

1. Weather is synthetic and based on approximate monthly normals; real TMY files and morphed future files are required.
2. The 1R1C model ignores zoning, radiant asymmetry and detailed airflow; multi-zone EnergyPlus models are needed.
3. Daylight, glare and circadian indicators are proxies; Radiance-based simulation should replace them.
4. The well-being index depends on weights, which here are illustrative.
5. Rule-based dispatch underestimates achievable autonomy compared with optimised control.
6. Qualitative criteria are only included in the web tool.

7.3 Validation pathway

The full programme should validate the workflow against monitored Greek buildings, validate surrogates against held-out EnergyPlus runs using ASHRAE Guideline 14 criteria, and test the digital tool with design practitioners to measure whether it changes early decisions.

8 Conclusions

Integrating energy autonomy and well-being in early design requires hourly energy balance, a broad computable well-being indicator set and fast surrogates, all built around the facade as a shared resource. Applied to 69 Greek locations, a simplified demonstrator shows that (i) annual net-positive balances still leave 34–53% of demand imported; (ii) storage and roof PV dominate autonomy, while glazing, insulation and form dominate well-being, and insulation benefits both; (iii) the two objectives conflict only near the Pareto front; (iv) occupancy timing,

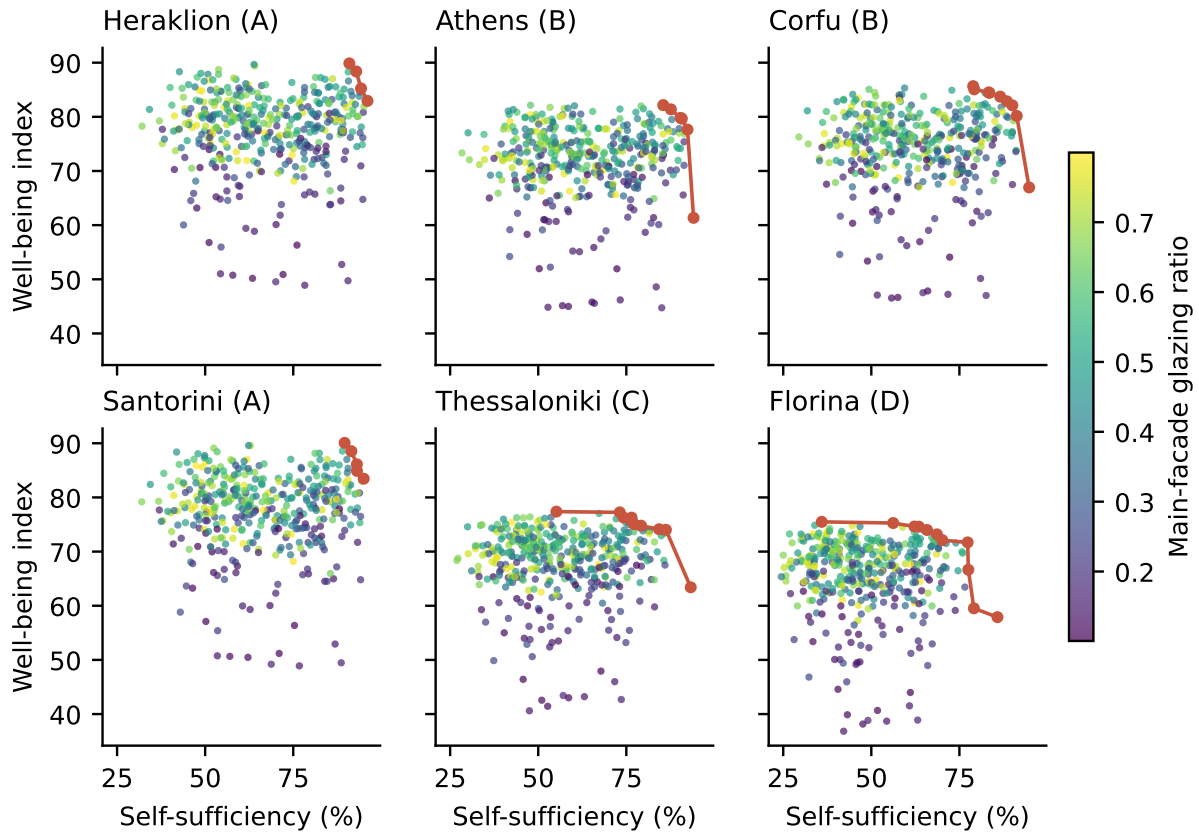


Figure 8: Self-sufficiency versus well-being for 400 designs per location, coloured by main-facade glazing ratio; red lines mark the Pareto front.

notably seasonal tourism, strongly affects autonomy; (v) warming improves autonomy slightly but degrades passive comfort in the south; and (vi) a climate-aware surrogate generalises to unseen locations.

Future work will replace the demonstrator with the full EnergyPlus/Radiance/pvlib workflow and morphed weather files, elicit well-being weights from occupants and experts, train hybrid surrogates on hourly profiles, add optimised dispatch and heat-pump flexibility, and extend the typologies to hotels and schools.

Data and tool availability

The demonstrator is implemented as an open web tool covering all 69 locations, with interactive design parameters, hourly balance maps, well-being indicators, design-space sampling, Pareto analysis and an on-the-fly surrogate. The tool is distributed as a self-contained package that runs in any modern web browser without installation.

References

- AIVC. EPBD 2024 recast in relation to IEQ and ventilation. <https://www.aivc.org/content/epbd-2024-recast-relation-ieq-and-ventilation>, 2026.
- ASHRAE. Guideline 14: Measurement of energy, demand, and water savings, 2014.

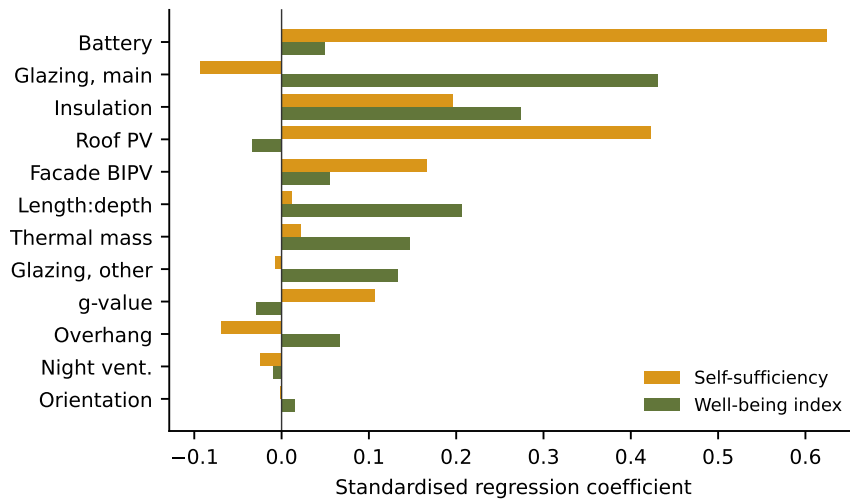


Figure 9: Standardised regression coefficients of design parameters on self-sufficiency and well-being (pooled over six locations).

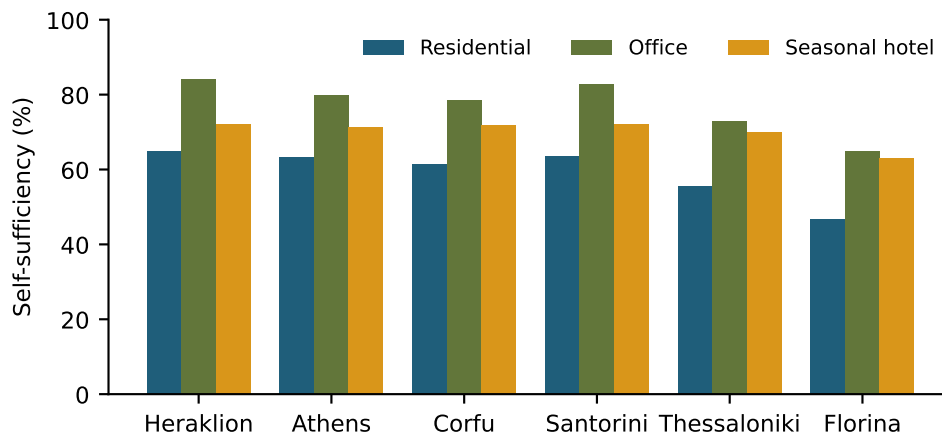


Figure 10: Self-sufficiency of the reference building under three use patterns.

BUILD UP. Acceptable indoor environmental quality and energy efficiency standards: can buildings meet both (and for everyone)? <https://build-up.ec.europa.eu/en/resources-and-tools/articles/acceptable-indoor-environmental-quality-and-energy-efficiency>, 2025.

CEN. EN 17037: Daylight in buildings, 2018.

CEN. EN 16798-1: Energy performance of buildings – ventilation for buildings – part 1: Indoor environmental input parameters, 2019.

CIBSE. TM52: The limits of thermal comfort – avoiding overheating in European buildings, 2013.

CIE. CIE S 026/E:2018 system for metrology of optical radiation for ipRGC-influenced responses to light, 2018.

Clean Energy for EU Islands. Net-metering / net-billing in Greece. <https://clean-energy-islands.ec.europa.eu/countries/greece/legal/electricity-support/net-metering-net-billing-net-metering-and-virtual-net>, 2025.

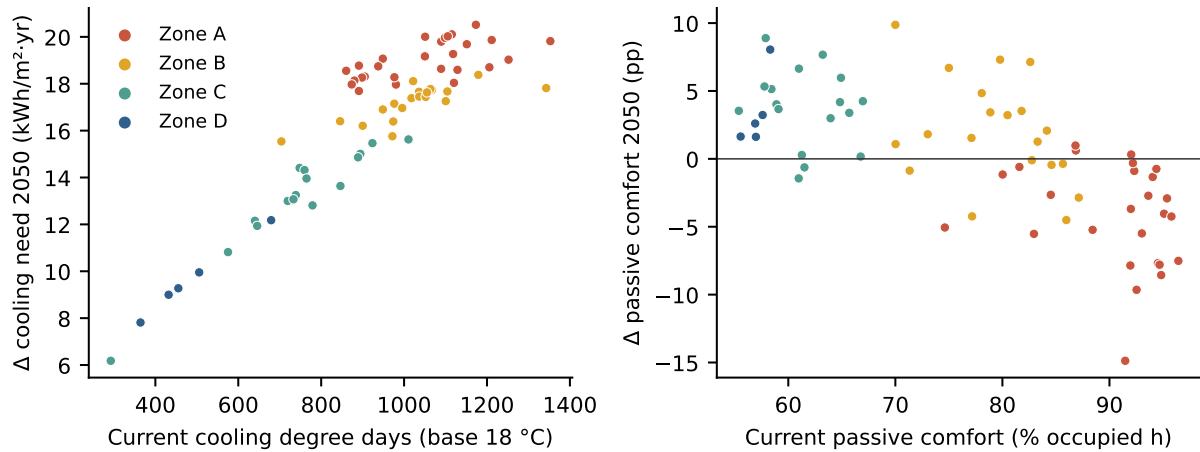


Figure 11: Change in cooling need versus current cooling degree days (left) and change in passive comfort versus current comfort (right) under the warming scenario.

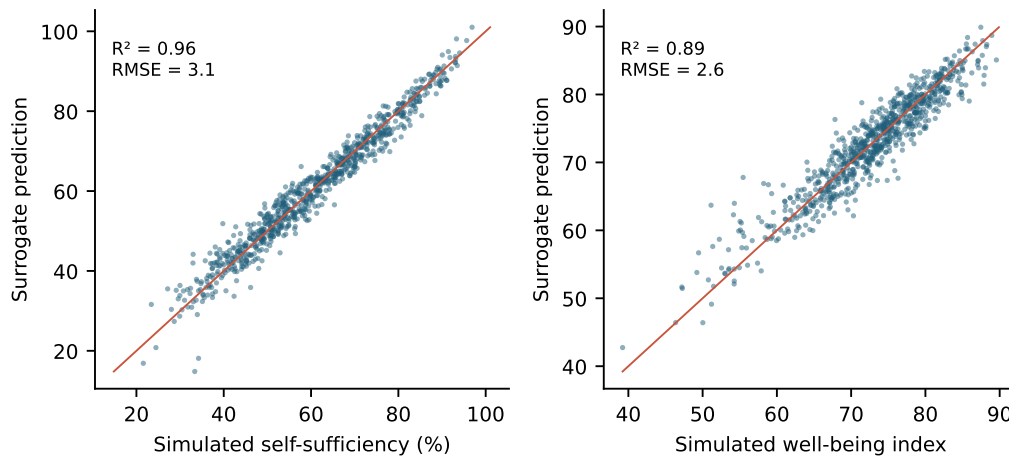


Figure 12: Parity plots for the cross-location surrogate on 14 held-out locations.

Raymond J. Cole. Transitioning from green to regenerative design. *Building Research & Information*, 40(1):39–53, 2012.

CORDIS, European Commission. A remote Greek island is on its way to becoming energy self-sufficient. <https://cordis.europa.eu/article/id/124099-a-remote-greek-island-is-on-its-way-to-becoming-energy-self-sufficient>, 2018.

Drury B. Crawley et al. EnergyPlus: creating a new-generation building energy simulation program. *Energy and Buildings*, 33(4):319–331, 2001.

Elena G. Dascalaki, Constantinos A. Balaras, Athina G. Gaglia, Kalliopi G. Droutsas, and Simon Kontoyiannidis. Energy performance of buildings—EPBD in Greece. *Energy Policy*, 45:469–477, 2012.

Richard de Dear and Gail S. Brager. Developing an adaptive model of thermal comfort and preference. *ASHRAE Transactions*, 104(1), 1998.

Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002.

- EPBD Concerted Action. Implementation of the EPBD in Greece: status in 2020. <https://cdn.ca-epbd.eu/epbd/Media/638373599786588364/Implementation-of-the-EPBD-in-Greece-2020.pdf>, 2020.
- D. G. Erbs, S. A. Klein, and J. A. Duffie. Estimation of the diffuse radiation fraction for hourly, daily and monthly-average global radiation. *Solar Energy*, 28(4):293–302, 1982.
- EUR-Lex. Energy performance of buildings: summary of directive (EU) 2024/1275. <https://eur-lex.europa.eu/EN/legal-content/summary/energy-performance-of-buildings.html>, 2024. Accessed 25 September 2026.
- Filippo Giorgi. Climate change hot-spots. *Geophysical Research Letters*, 33:L08707, 2006.
- Jon Herman and Will Usher. SALib: an open-source python library for sensitivity analysis. *Journal of Open Source Software*, 2(9):97, 2017.
- Lisa Heschong. *Thermal Delight in Architecture*. MIT Press, Cambridge, MA, 1979.
- William F. Holmgren, Clifford W. Hansen, and Mark A. Mikofski. pvlib python: a python package for modeling solar energy systems. *Journal of Open Source Software*, 3(29):884, 2018.
- ICE. Using climate risk analytics to plan for hotter European summers. <https://www.ice.com/insights/climate-risk/using-climate-risk-analytics-to-plan-for-hotter-european-summer>, 2026.
- ISO. ISO 13790:2008 energy performance of buildings – calculation of energy use for space heating and cooling, 2008.
- Mark F. Jentsch, AbuBakr S. Bahaj, and Patrick A. B. James. Climate change future proofing of buildings: generation and assessment of building simulation weather files. *Energy and Buildings*, 40(12):2148–2168, 2008.
- Athanasios F. Karagiannidis, Kostas Lagouvardos, Vassiliki Kotroni, and Elissavet Galanaki. Expected changes in heating and cooling degree days over Greece in the near future based on climate scenarios projections. *Atmosphere*, 15(4):393, 2024. doi: 10.3390/atmos15040393.
- George Katavoutas, Dimitra Founda, Gianna Kitsara, and Christos Giannakopoulos. Climate change and thermal comfort in top tourist destinations—the case of Santorini (Greece). *Sustainability*, 13:9107, 2021. doi: 10.3390/su13169107.
- Lexology. The self-consumption regime in Greece. <https://www.lexology.com/library/detail.aspx?g=44b0972a-ba9e-41b3-bfc8-550d22b25db7>, 2024.
- Paul Littlefair. Site layout planning for daylight and sunlight: a guide to good practice. BRE Report BR 209, 2nd ed., 2011.
- Longevity Partners. New buildings under the revised EPBD: from nearly zero-energy to zero-emission. <https://longevity-partners.com/exclusive/new-buildings-under-the-revised-epbd-from-nearly-zero-energy-to-zero-emission/>, 2026.
- Robert J. Lucas et al. Measuring and using light in the melanopsin age. *Trends in Neurosciences*, 37(1):1–9, 2014.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30, 2017.

- Rasmus Luthander et al. Photovoltaic self-consumption in buildings: a review. *Applied Energy*, 142:80–94, 2015.
- J. Fergus Nicol and Michael A. Humphreys. Adaptive thermal comfort and sustainable thermal standards for buildings. *Energy and Buildings*, 34(6):563–572, 2002.
- Norton Rose Fulbright. Securing sustainable energy independence for the Greek islands: the role of storage. <https://www.nortonrosefulbright.com/en/knowledge/publications/60e9ad14/securing-sustainable-energy-independence-for-the-greek-islands-the-role-of-storage>, n.d.
- Torben Østergård, Rasmus L. Jensen, and Steffen E. Maagaard. Building simulations supporting decision making in early design: a review. *Renewable and Sustainable Energy Reviews*, 61: 187–201, 2016.
- Vasileios Panousis. Climate zones of the hellenic territory (revised KENAK 2017). <https://www.district-energy.gr/energy/energy-building-efficiency/climate-zones-greece/>, 2018. Accessed 25 September 2026.
- Bill Reed. Shifting from ‘sustainability’ to regeneration. *Building Research & Information*, 35(6): 674–680, 2007.
- Christoph F. Reinhart, John Mardaljevic, and Zack Rogers. Dynamic daylight performance metrics for sustainable building design. *LEUKOS*, 3(1):7–31, 2006.
- Eugénio Rodrigues, Marco S. Fernandes, and David Carvalho. Future weather generator for building performance research: an open-source morphing tool and an application. *Building and Environment*, 233, 2023.
- Mostapha Sadeghipour Roudsari and Michelle Pak. Ladybug: a parametric environmental plugin for grasshopper to help designers create an environmentally-conscious design. In *Proceedings of Building Simulation 2013*, Chambéry, France, 2013.
- Jaume Salom et al. Understanding net zero energy buildings: evaluation of load matching and grid interaction indicators. In *Proceedings of Building Simulation 2011*, Sydney, Australia, 2011.
- Igor Sartori, Assunta Napolitano, and Karsten Voss. Net zero energy buildings: a consistent definition framework. *Energy and Buildings*, 48:220–232, 2012.
- Technical Chamber of Greece. T.O.T.E.E. 20701-3/2010: Climatic data of hellenic regions (table of prefectures by climate zone). <http://portal.tee.gr/portal/page/portal/tptee/totee/TOTEE-20701-3-Final-TEE%202nd.pdf>, 2010.
- Eleftheria Touloupaki and Theodoros Theodosiou. Optimization of external envelope insulation thickness: a parametric study. *Energies*, 10(3):270, 2017. doi: 10.3390/en10030270.
- Ilias Tsagas. Making Greece’s islands energy self sufficient. pv magazine, <https://www.pv-magazine.com/2024/12/20/making-greeces-islands-energy-self-sufficient/>, 2024.
- Ilias Tsagas. Greece installs 400 MW of net-metered solar in 2024. pv magazine, <https://www.pv-magazine.com/2025/01/31/greece-adds-400-mw-of-solar-under-net-metering-in-2024/>, 2025.
- Paul Westermann and Ralph Evins. Surrogate modelling for sustainable building design: a review. *Energy and Buildings*, 198:170–186, 2019.

Jan Wienold and Jens Christoffersen. Evaluation methods and development of a new glare prediction model for daylight environments with the use of CCD cameras. *Energy and Buildings*, 38(7):743–757, 2006.

A Location data and results

Table 5: All 69 locations: KENAK zone, coordinates, synthetic-climate indicators and reference-design results (today and 2050). SS = self-sufficiency, PC = passive comfort, WB = well-being index.

Location	Region	Zone	Lat (°N)	Lon (°E)	HDD (K d)	CDD	GHI (kWh/m ²)	SS (%)	PC (%)	PC ₂₀₅₀ (%)
Agios Nikolaos	Crete	A	35.19	25.72	571	1211	1893	66	93	83
Chania	Crete	A	35.51	24.02	746	1050	1883	64	92	88
Heraklion	Crete	A	35.34	25.13	645	1090	1891	65	95	87
Ierapetra	Crete	A	35.01	25.74	519	1353	1893	66	91	77
Rethymno	Crete	A	35.37	24.47	615	1051	1890	65	96	89
Sitia	Crete	A	35.21	26.10	565	1099	1893	65	95	87
Milos	Cyclades	A	36.73	24.43	659	938	1823	64	96	92
Mykonos	Cyclades	A	37.45	25.33	782	891	1788	63	92	91
Naxos	Cyclades	A	37.10	25.38	777	891	1805	63	94	93
Paros	Cyclades	A	37.08	25.15	743	860	1806	63	94	94
Santorini	Cyclades	A	36.42	25.43	709	949	1839	64	95	91
Syros (Ermoupoli)	Cyclades	A	37.44	24.94	732	899	1789	63	94	91
Astypalaia	Dodecanese	A	36.55	26.35	729	905	1832	64	95	92
Karpathos	Dodecanese	A	35.51	27.21	536	1173	1883	65	95	86
Kos	Dodecanese	A	36.89	27.29	714	1151	1815	65	92	84
Rhodes	Dodecanese	A	36.43	28.22	613	1115	1838	65	95	87
Tilos	Dodecanese	A	36.43	27.37	671	1105	1838	64	93	88
Argostoli (Kefalonia)	Ionian Islands	A	38.18	20.49	798	980	1735	62	87	87
Ithaca	Ionian Islands	A	38.37	20.72	842	977	1726	62	87	88
Kythira	Ionian Islands	A	36.15	22.99	832	874	1852	63	92	92
Zakynthos	Ionian Islands	A	37.78	20.90	754	1118	1754	64	88	83
Ikaria	North & East Aegean	A	37.62	26.17	799	880	1780	63	92	92
Samos	North & East Aegean	A	37.75	26.98	904	1089	1774	62	82	81
Kalamata	Peloponnese	A	37.04	22.11	850	1129	1773	63	85	82
Nafplio	Peloponnese	A	37.57	22.80	917	1120	1748	62	80	79
Sparti	Peloponnese	A	37.07	22.43	1040	1206	1770	60	75	70
Aegina	Saronic Islands	A	37.75	23.43	796	1252	1774	64	83	77
Athens	Attica	B	37.98	23.73	958	1343	1727	63	77	73
Piraeus	Attica	B	37.94	23.65	852	1179	1730	63	86	81
Chalkida	Central Greece	B	38.46	23.60	1043	1101	1704	62	77	79
Lamia	Central Greece	B	38.90	22.43	1235	1104	1683	59	70	71
Livadeia	Central Greece	B	38.44	22.88	1366	972	1705	58	71	70
Arta	Epirus	B	39.16	20.99	1064	974	1671	60	79	82
Igoumenitsa	Epirus	B	39.50	20.27	1213	900	1655	59	75	82
Preveza	Epirus	B	38.96	20.75	1009	949	1680	61	83	90
Corfu	Ionian Islands	B	39.62	19.92	1019	976	1667	61	80	84
Lefkada	Ionian Islands	B	38.83	20.71	928	1022	1704	62	83	85
Paxos	Ionian Islands	B	39.20	20.18	881	995	1687	63	85	84
Chios	North & East Aegean	B	38.37	26.14	899	1067	1743	63	87	84
Myrina (Lemnos)	North & East Aegean	B	39.87	25.06	1383	704	1670	58	70	80
Mytilene (Lesvos)	North & East Aegean	B	39.11	26.55	987	1018	1707	62	82	85
Corinth	Peloponnese	B	37.94	22.93	942	1064	1730	62	83	83
Patras	Peloponnese	B	38.25	21.73	963	1036	1714	62	84	86
Pyrgos	Peloponnese	B	37.68	21.44	956	1037	1742	62	86	85
Skiathos	Sporades	B	39.16	23.49	1085	846	1704	61	80	87

Location	Region	Zone	Lat	Lon	HDD	CDD	GHI	SS	PC	PC ₂₀₅₀
Volos	Thessaly	B	39.36	22.94	1210	1052	1661	60	73	75
Agrinio	Western Greece	B	38.62	21.41	1090	1055	1697	60	78	83
Karpenisi	Central Greece	C	38.91	21.80	2433	292	1668	48	55	59
Ioannina	Epirus	C	39.66	20.85	1764	646	1632	54	67	71
Edessa	Macedonia	C	40.80	22.05	1978	575	1584	51	58	67
Katerini	Macedonia	C	40.27	22.50	1724	733	1618	54	65	71
Kavala	Macedonia	C	40.94	24.41	1672	760	1597	56	65	69
Kilkis	Macedonia	C	40.99	22.87	2023	719	1597	52	58	64
Polygyros (Chalkidiki)	Macedonia	C	40.38	23.44	1690	748	1613	55	66	69
Serres	Macedonia	C	41.09	23.55	1909	846	1597	53	58	63
Thessaloniki	Macedonia	C	40.64	22.94	1731	924	1601	55	61	60
Veria	Macedonia	C	40.52	22.20	1843	739	1606	53	61	68
Tripoli	Peloponnese	C	37.51	22.37	1723	640	1733	54	63	71
Karditsa	Thessaly	C	39.36	21.92	1614	889	1661	56	64	67
Larissa	Thessaly	C	39.64	22.42	1702	1011	1649	55	62	61
Trikala	Thessaly	C	39.56	21.77	1685	894	1653	55	67	67
Alexandroupoli	Thrace	C	40.85	25.87	1894	765	1597	54	61	62
Komotini	Thrace	C	41.12	25.41	1940	779	1597	52	59	63
Xanthi	Thrace	C	41.13	24.89	1953	733	1597	52	59	63
Drama	Macedonia	D	41.15	24.15	2100	679	1597	52	58	66
Florina	Macedonia	D	40.78	21.41	2629	364	1584	47	56	57
Grevena	Macedonia	D	40.08	21.43	2417	455	1612	49	57	59
Kastoria	Macedonia	D	40.52	21.26	2443	432	1592	48	57	60
Kozani	Macedonia	D	40.30	21.79	2337	506	1602	49	58	61